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Key Points:

- Natural hydrogen exploration in former oil and gas province
- Use of the Optical Character Recognition algorithm to optimize processing of a large drilling report database
- Indices for a potential H₂ system (source, migration, trap) in the Paris Basin

Supporting Information:

Supporting Information may be found in the online version of this article.

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Characterizing Natural Hydrogen Occurrences in the Paris Basin From Historical Drilling Records

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Abstract This study investigates natural hydrogen (H₂) occurrences in the Paris Basin, using Optical Character Recognition (OCR) technology to analyze an extensive, yet underexploited, database that contains historic drilling records. The potential of natural hydrogen has been largely unexplored in conventional oil and gas wells. Utilizing the in-house CVAGeoDB database based on public well data, which includes well logs, mudlogs, and End Drilling Reports (EDRs) in PDF image format, we applied the Tesseract-OCR Engine to convert these documents into searchable formats for efficient data analysis. Our analysis revealed several H₂-bearing wells across French sedimentary basins. The hydrogen occurrences in the Aquitaine Basin may be explained by the geological context and in particular the presence of a mantle body at shallow depth. On the contrary, the detection of H₂ in the Paris Basin cannot be explained in a straightforward manner as the presence of ultramafic or U-rich rocks is poorly documented so far. In the Paris Basin, H₂ has been detected in four main formations: the Lusitanian, the Dogger, and Triassic aquifers as well as in the basement. The highest hydrogen concentration (52 vol%) was measured in the Dogger aquifer. These wells are primarily located along the Bray Fault, indicating at least a structural influence on H₂ distribution. Finally, the presence of serpentinized dunite from the Lizard complex associated with the bedrock may have played the role as a source for H₂. This research demonstrates the effectiveness of OCR in reassessing historical drilling data for natural hydrogen exploration, highlighting the need for comprehensive exploration methodologies in this emerging field.

Plain Language Summary This study explores the presence of natural hydrogen (H₂) in the Paris Basin, employing Optical Character Recognition (OCR) technology to sift through an extensive database of old drilling records that have not been fully utilized in the past. As the world increasingly seeks carbon-neutral energy sources, natural hydrogen produced through interactions between water and rocks emerges as a promising alternative to fossil fuels. Our research focuses on the CVAGeoDB database, which contains detailed information on drilling activities but in a non-searchable PDF image format. OCR is a tool that turns images containing text, such as scanned documents, into text files that one can easily search and analyze. Our findings indicate the presence of H₂ in several wells across French sedimentary basins. The Paris Basin, exhibits unexpected H₂ occurrences not directly linked to anticipated geological factors classically used in H₂ exploration. In the Paris Basin, the highest hydrogen concentration (52 vol%) was discovered in the Dogger aquifer. These wells are predominantly situated along the Bray Fault, suggesting a structural control on the distribution of hydrogen. This research underscores the utility of OCR in re-evaluating historical drilling data for natural hydrogen exploration.

1. Introduction

Geochemical interactions between water and rock are known to generate molecular hydrogen (H₂) under specific conditions. These processes, extensively documented in the literature (e.g., Klein et al., 2020; Milkov, 2022), include the serpentinization of ultramafic rocks, the hydration of ferrous minerals like siderite or amphibole, the radiolysis of water, and the diagenesis of organic matter. These geochemical processes are not only crucial for understanding Earth's deep microbial ecosystems but also play a significant role in the abiotic synthesis of organic molecules (Lin, Slater, et al., 2005; Truche et al., 2020).

In light of the global shift toward sustainable energy sources, naturally occurring H₂ has garnered significant attention as a putative carbon-neutral energy alternative to traditional fossil fuels. This interest is reflected in recent research exploring the potential of H₂ in various geological contexts (Donzé et al., 2020; Moretti et al., 2021; Prinzhofner et al., 2018; N. J. P. Smith et al., 2005; Truche et al., 2024). Current exploration methodologies for

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targeting H₂ in potential geological reservoirs predominantly rely on soil gas analysis at a depth of approximately 1 m. However, this approach is somewhat limited in scope and does not encompass a comprehensive and efficient methodology. This gap in methodology underscores the need for more integrated and systematic exploration strategies, as highlighted in recent studies (Lefeuvre et al., 2021, 2022; McMahon et al., 2023).

In the domain of petroleum geology, the presence of molecular hydrogen in natural reservoirs has historically been underappreciated (Milkov, 2022). This oversight can be attributed to the fact that H₂ was rarely detected in the multitude of wells drilled globally for oil and natural gas exploration, as noted by Gaucher (2020). The standard geochemical sensors employed in these wells were primarily calibrated for detecting fossil hydrocarbons, such as methane, with less sensitivity or focus on H₂.

However, emerging research and field data have begun to challenge this long-standing viewpoint. Recent studies have identified significant concentrations of H₂ in various geological settings across the world. In particular, the following notable examples can be mentioned. The “Bougou-1” water exploration well, which was drilled in 1987 in Mali, revealed H₂ concentration of 98 vol% at the well head (Maiga et al., 2023). The “Tisovita geothermal well,” which was drilled in the 1970’ in Romania, shown H₂ concentration as high as 28.7 vol% (Mitrofan et al., 2021). The “#1 Wilson” hydrocarbon exploration well, drilled in 2006 in the Kansas (USA), displayed H₂ concentration of 10.0 vol% (Newell et al., 2007). A concentration of 42.7 vol% H₂ was reported from the mineral exploration borehole EVDD008 drilled in the Yilgarn Craton in Australia (Boreham et al., 2021). At the “Copper Cliff” mineral exploration borehole in Canada, the H₂ concentration ranged from 9.9 to 57.8 vol% (Sherwood et al., 1988).

These findings underscore the potential for natural hydrogen reservoirs and highlight the necessity to refine and adapt the analytical techniques used for prospecting energy and mineral resources, particularly for the detection and quantification of H₂. The implications of these discoveries are significant not only for understanding subsurface geochemical processes but also for evaluating the potential of H₂ as an energy resource in the context of a transitioning global energy landscape.

In France, the exploration and production of oil and gas have led to the drilling of over 5,000 wells. However, none of these wells have been designed for natural hydrogen exploration. The in-house CVAGeoDB database used in this study is composed of well logs, mudlogs and End Drilling Reports (EDRs) which are in PDF image format. Manual examination of each well would take a considerable amount of time, so an Optical Character Recognition (OCR) algorithm (Tesseract-OCR Google Engine; R. Smith, 2007) was used to transform these scanned image PDFs into searchable PDFs.

The primary objective of this study is to demonstrate the efficacy of applying OCR technology to old but extensive drilling datasets. This approach enables the rapid identification of unexpected but promising areas for H₂ exploration. The Paris Basin, with its unique geological features, serves as a focal point for this investigation, potentially positioning it as a H₂-rich geological province. Our research underscores the value of innovative data processing techniques in enhancing the efficiency and scope of geological exploration, particularly in the context of emerging energy resources such as natural hydrogen.

2. Geological Context

The Paris Basin, a prominent intracratonic basin, encompasses approximately 3 km of sedimentary deposits spanning from the Triassic to the present. These deposits overlie the Variscan suture zones, a significant geological feature from the Upper Paleozoic era, as documented in several studies (Figure 1a; Curnelle & Dubois, 1986; Delmas et al., 2002; Megnien, 1980; Pomerol, 1978). The basin is geographically bounded by four Paleozoic massifs: the Armorican Massif to the south, the Bresse Graben to the southeast, the Vosges Massif to the east, and the Ardennes Massif to the north. The Paris Basin exhibits a distinct geological structure characterized by concentric outcrops of Meso-Cenozoic rocks. This structure is the result of a series of depositions and erosional processes, as detailed in various studies (Beccaleto et al., 2011; Delmas et al., 2002; Guillocheau et al., 2000). Additionally, the basin extends to connect with the London and Belgium Basins to the northwest and north, respectively (Dercourt et al., 2000).

Two major fault systems are prominent in the Paris Basin: the Sennely fault and the Bray Fault system (Figure 1). The latter is a N130° dextral strike-slip fault that impacts the sedimentary cover (Matte & Hirn, 1988; Raoult & Meilliez, 1987).

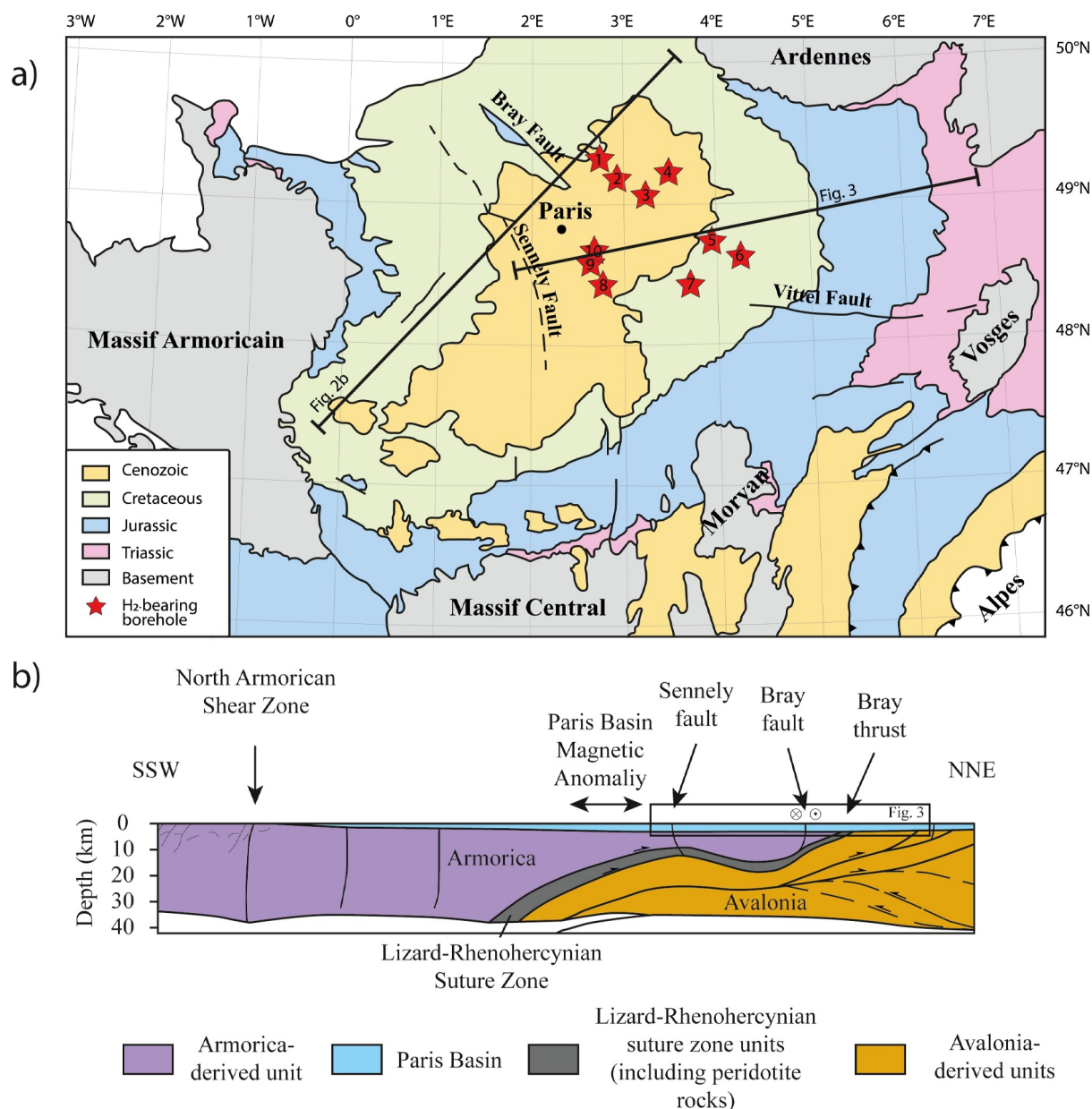


Figure 1. (a) The structural map of the Paris Basin displays the main units and the surrounding crystalline massifs (modified from Baptiste (2016)). In this map, the red stars represent the wells showing evidence of the presence of H₂, as identified using the Optical Character Recognition algorithm developed in the study 1—Longueil; 2—Betz; 3—Montreuil Aux Lions; 4—Cramaille; 5—Connantre 1 and 2; 6—Grandville; 7—Saint Martin de Bossenay; 8—Hericy; 9—Le Luteau; 10—Coubert. (b) Crustal-scale cross-section through the Variscan orogenic system and the Paris Basin (modified from Averbuch and Piromallo (2012) and Matte and Hirn (1988)).

The basement of the Paris Basin is a lithological and structural inheritance of the Variscan orogeny, which occurred during the Carboniferous period following the north-south convergence of the Avalonia and Gondwana plates, culminating in the closure of the Rheic Ocean (Averbuch & Piromallo, 2012; Ballèvre et al., 2009; Matte, 1986). Seismic tomography beneath the Paris Basin has revealed anomalies in the upper mantle, with V_p velocities oriented along NW-SE axes. These anomalies, situated along the Bray Fault, are associated with the Variscan suture zone, indicative of a Variscan paleoslab (Figure 1b; Averbuch & Piromallo, 2012; Cazes et al., 1986). Recent P-wave seismic tomography studies suggest the presence of a subducted paleo-slab beneath a segment of the Bray Fault (Autran et al., 1994; Averbuch & Piromallo, 2012; Matte & Hirn, 1988). Complementary gravimetric and magnetic data along this major fault structure have identified anomalies correlating with granite intrusions (Baptiste, 2016; Thébaud et al., 2006). The Lizard-Rhenohercynian suture zone, as illustrated

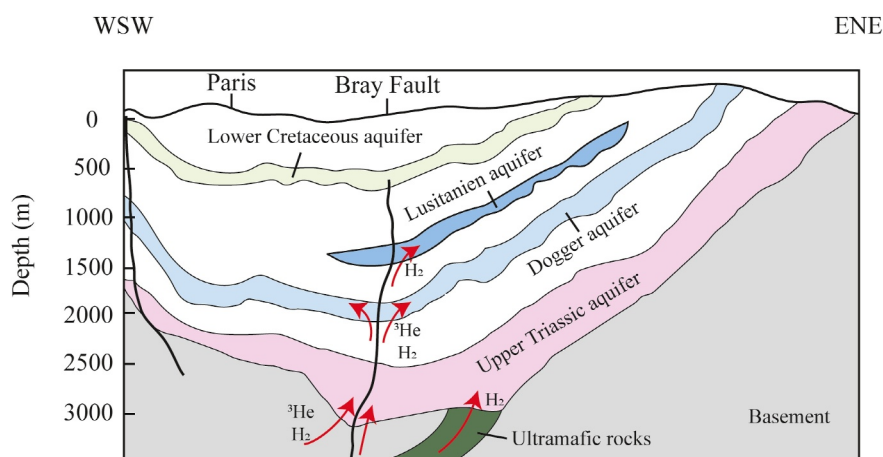


Figure 2. A schematic cross-section of the Paris Basin, with the positions of the four main aquifers where H_2 was detected and the putative H_2 system beneath the Paris Basin (Modified from Pinti and Marty (1998)).

by the Lizard ophiolitic complex in the southern part of Great Britain, comprises both ultramafic (peridotite, serpentinite) and crustal rocks (amphibolite, gneiss, etc.), as further elucidated in various studies (Cook et al., 1998, 2002; Leake & Styles, 1984; Roberts et al., 1993).

The Paris Basin, primarily recognized for its hydrocarbon reserves, is characterized by over 3,000 boreholes that have been instrumental in delineating its structural framework. This basin encompasses several key aquifers and reservoirs (Figure 2). These include: (a) The Upper Triassic Sandstone aquifer, capped by mudstone and evaporitic rocks, particularly in the central part of the basin with a temperature of 120°C (e.g., Montmirail well; Torelli et al., 2020); (b) The Middle Jurassic (hereafter “Dogger”) aquifer is a 160 m thick oolitic limestone reservoir located between Lower Jurassic and Lower Callovian marls (Bril et al., 1994). This aquifer contains geothermal waters with temperatures ranging between 50 and 80°C and (c) The Upper Jurassic (hereafter “Lusitanian”) made of limestone; (d) The Lower Cretaceous green sandstone aquifers. A recent study calculated the geothermal gradient at $T = 10 + 34.9 \times z$, (where T is the temperature in $^\circ\text{C}$ and z is the depth in km), derived from an analysis of existing well data and aquifer temperatures (Bonté et al., 2010; Pinti & Marty, 1998).

Furthermore, Pinti and Marty (1998) have identified an helium enrichment anomaly in the Dogger, characterized by an unusual excess of mantle-derived ^3He in the Dogger aquifer, particularly near the Bray Fault (Figure 2). This fault is hypothesized to serve as a conduit for helium-rich fluids, facilitating their migration through the low-permeability shales that separate the Triassic and Dogger aquifers (Worden & Matray, 1995).

By integrating geodynamic, geophysical, and geochemical data, it was proposed that a paleo-slab, composed of mantellic and metamorphic rocks, is connected to the basin via the Bray Fault (Bril et al., 1994). The detection of mantle-derived ^3He in the Dogger aquifer further suggests a deep structural connection to these formations (Figure 2).

The Paris Basin's geology is particularly promising for H_2 exploration. This potential is attributed to the presence of ultramafic rocks undergoing serpentinization, water radiolysis in granite bodies, the existence of preferential pathways for fluid migration and the excess of ^3He . Additionally, numerous reservoirs within the basin exhibit favorable porosity and permeability characteristics, making them suitable candidates for hydrogen storage. Collectively, these factors underscore the basin's potential as a complete H_2 system from sources to traps.

3. Methodology

3.1. Data Origin

The database used in our research encompasses a comprehensive collection of public well data archived by CVA Group®. This in-house repository, known as the CVAGeoDB database, contains data from 3,246 wells covering a part of hydrocarbon exploration activities in both metropolitan France and its overseas territories. It primarily consists of End Drilling Reports (EDRs) in PDF format. This dataset well distributed across the territory is

representative of the 5,139 exploration wells drilled in France since 1927. The database incorporates a diverse array of data, including scanned EDRs, well logs (such as Gamma ray or bulk density), mudlogs, seismic boreholes, and other relevant documents (e.g., site logs, tests). For the purposes of our study, we concentrated exclusively on the analysis of the EDRs, which comprise 3,246 scanned documents in PDF format, with individual files ranging from 10 to 300 pages. Due to the lack of standardization in Environmental Data Records (EDRs), the detailed methodology for gas analysis is often not elucidated.

3.2. Pytesseract Screening RFS

Given the image-based nature of these documents, conventional keyword search techniques are infeasible, and manual examination would be prohibitively time-consuming. To address this challenge, we employed OCR technology. Specifically, we utilized Pytesseract, a Python-based OCR tool that integrates Google's Tesseract-OCR Engine (available at <https://github.com/madmaze/pytesseract>), to convert the EDR PDFs into searchable formats (Figure S1 in Supporting Information S1).

This process requires preliminary processing of each EDR. As Pytesseract is incapable of directly processing PDF files, we first converted each page of the EDRs into image files (PNG, JPG, etc.) using an initial conversion tool. This tool transforms the scanned PDF images into Python Imaging Library (PIL) format, with each PIL image representing a page from the EDR. Subsequently, Pytesseract OCR processes each page, converting the text into a searchable PDF format, which we labeled as “EDR_reference.”

To further analyze the newly created searchable database, we employed PyMuPDF, a Python library designed for extracting, analyzing, converting, and manipulating PDF file data. This library facilitates keyword filtering within each searchable PDF, allowing us to extract the file name and the specific page number where the keyword appears, and store this information in a.txt file. This approach significantly streamlines the process of identifying relevant files and their locations, thereby enhancing the efficiency of quality checks and validation. In our case study, we focused on keywords such as “H₂,” “Hydrogène.”

4. Results

Applying the OCR algorithm to the database, 141 occurrences were found in the CVAGeoDB database containing “hydrogen” (“hydrogène” in French) and/or “H₂.” Initially, each time the word “hydrogen” is detected, *Pytesseract* records it in the results file. However, in the boreholes of the Paris Basin, there are many occurrences of Hydrogen Sulfide (H₂S), which biases the results. The inability to distinguish between “H₂” and “H₂S” can lead to problems; therefore, rapid human verification is essential. The main results and the detailed well names and numbers have been reported in Supporting Information S1.

After the validation process, 26 wells (0.5% of the database) with H₂ detection have been identified in France, indicating the presence of H₂ during drilling operations or formation tests, with 11 of these wells situated in the Paris Basin (Figure 3, Tables S1 and S2). This limited number of H₂ detections might be indicative of under-reporting in other wells. Some drillings, such as a drilling EDR near Saint-Girons (Table S1), were equipped to measure H₂ but did not include these measurements in their final reports. Although previous drillings had the ability to detect H₂, this information was not always reported. Therefore, it is plausible that other drillings passed through H₂-rich zones without this being documented, leaving us uncertain about the actual presence of H₂ in some areas.

In EDR, when hydrogen is detected, geologists propose various hypotheses to explain its presence, such as: (a) hydrogen arising from tool deterioration, (b) hydrogen of unknown origin, (c) hydrogen detected but not quantified, (d) trace concentrations of hydrogen in the background gas, and (e) hydrogen identified using neutron porosity techniques. In this latter case, it is the hydrogen atom, mostly from water and clays, eventually from CH₄ that is detected and not H₂ as a high concentration/density of this molecule would be required to significantly impact the neutron signal.

Hydrogen was documented in the following wells.

- The Longueuil 1 well (drilled in 1972) detected H₂ and N₂ trace (no concentrations reported in the EDR) during mud gas logging at 3 different depths (a) In the limestone Lusitanian aquifer at 988–1,241 m (porosity of 19.2 %); (b) in the Dogger Massigny Marles at 1,241–1,347 m (porosity ranging from 9% to 19 %); and (c) In the

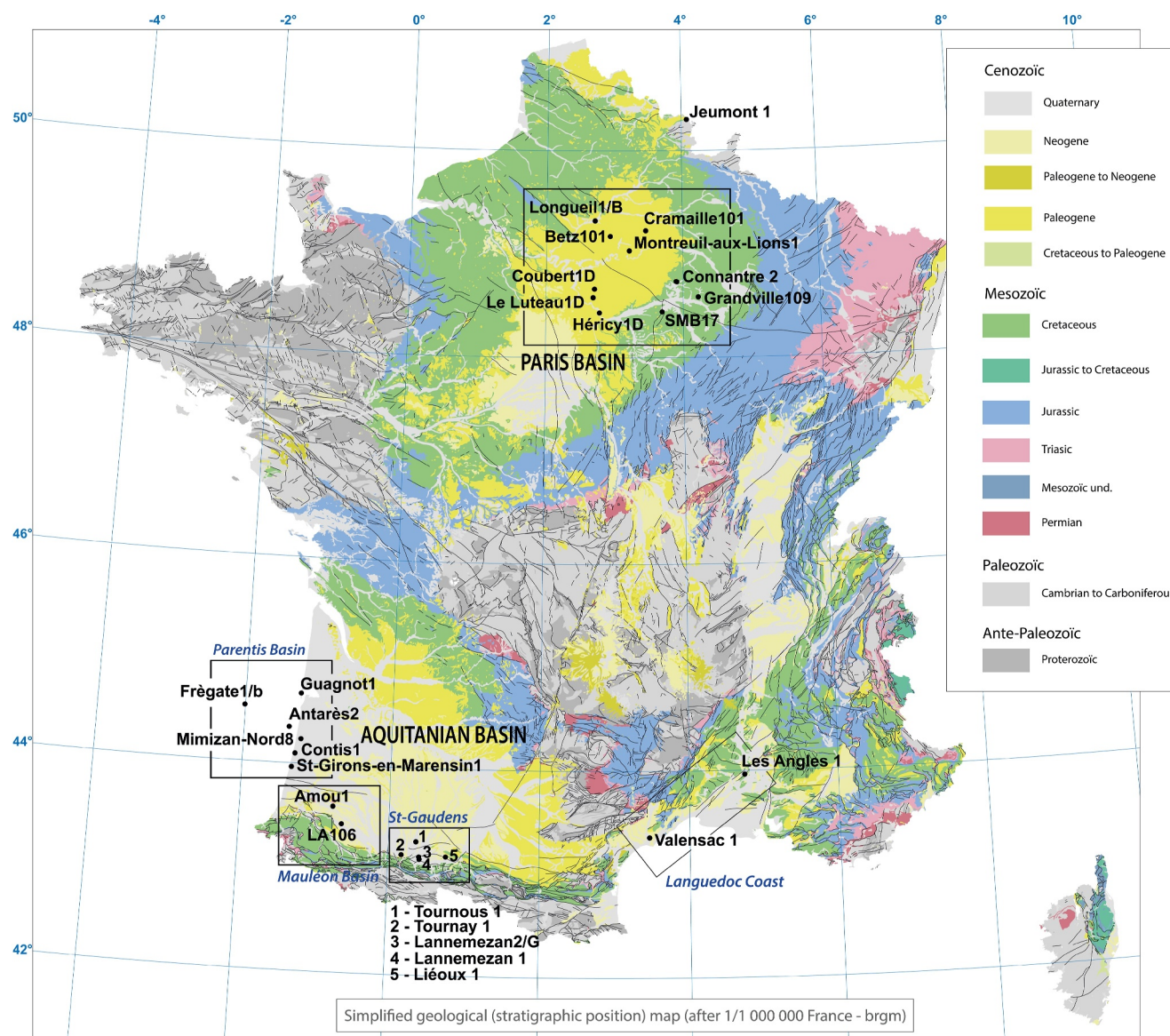


Figure 3. Geological map of France (1/1M—BRGM) highlighting the locations of wells where H_2 occurrences were detected using Optical Character Recognition.

Triassic Saint Maur Red Clay at 1,916–1,981 m. The EDR makes no mention of hydrocarbons when crossing these formations and doesn't explain the origin of the H_2 (Text S1 in Supporting Information S1).

- The Betz 101 well (drilled in 1963) revealed a concentration of H_2 gas ranging from 3 to 6 vol% in the Lusitanian limestone at a depth of 1,325–1,335 m; however, the report does not describe the methodology used for this gas analysis. The Lusitanian aquifer is mainly composed of limestone with a porosity ranging from 7% to 11%. The EDR makes no mention of hydrocarbons when crossing the Lusitanian formation and reveals that detecting high H_2 concentrations in the Lusitanian aquifer is common in this area, but its origin remains unknown (Text S2 in Supporting Information S1).
- The Montreuil Aux Lions 1 well (drilled in 1988) detected H_2 gas in a 1 L sample taken at a depth of 2,165–2,175 m corresponding to oolitic limestones in the Dogger. The pressure inside the bottle was 25 bar. The gas measurement revealed a gas composition of 52 vol% H_2 , 42 vol% CH_4 , and concentrations of C2, C3 and C4 alkanes of 4.3, 0.9, and 0.19 vol%, respectively. The EDR did not provide any information about the origin of this H_2 concentration (Text S3 in Supporting Information S1).
- The Cramaille 101 well (drilled in 1961) detected H_2 during mud gas logging while crossing the Lusitanian aquifer from 1,213 to 1,492 m depth, which is composed of limestones with a porosity of 16.7 % and a

- permeability ranging from 0.1 to 12 mD. However, the EDR did not indicate the exact concentration of H₂ gas, and there was no explanation provided for the presence of H₂ (Text S4 in Supporting Information S1). Moreover, the EDR makes no mention of hydrocarbons when crossing the Lusitanian formation
- The Connantre 2 well (drilled in 1981) detected the presence of H₂ and nitrogen from 1,508 to 1,533 m in the Dogger aquifer, which was composed of limestones with porosity ranging from 7% to 11%. The thickness of the effective reservoir was 9.5 m. The methodology for gas analysis is not described in the EDR (Text S5 in Supporting Information S1). The EDR mentions the presence of matrix and crack fluorescence with some hydrocarbons seepages on the core.
 - The Grandville 109 well (drilled in 1982) identified the presence of H₂ at two different depths during the mud gas logging. First, H₂ was detected in the Dogger aquifer that is made up of clayey limestones and ranges from 1,159 to 1,725 m depth. The background mud gases have a H₂ concentration from 0.25 to 0.65 vol%. Second, H₂ was also detected in the Triassic aquifer consisting of dolomite, clay, and evaporites from 2,053 to 2,555 m in depth. A concentration of 0.2 vol% of H₂ was detected in the gas retrieved from the drilling mud (Text S6 in Supporting Information S1).
 - The Coubert 1 well (drilled in 1986) detected H₂ during mud gas logging at the bottom hole at 2,547 m depth. However, no concentrations were reported in the EDR. Hercynian lithology at this depth is mainly composed of Gneissic rocks. The gas composition recorded showed unusual fluctuations, with a disappearance of C4 hydrocarbons and a reduction in C3 and C2 concentrations but an increase in total gas along with the presence of H₂. The main hypothesis proposed by the well geologists was a gas production linked to drill bit alteration (Text S7 in Supporting Information S1).
 - The Luteau 1 well (drilled in 1986) detected H₂ and CH₄ during mud gas logging while crossing Upper Triassic lithology from 2,569 to 2,578 m depths, which is mainly composed of clay but locally contains anhydrite. Unfortunately, the gas chromatograph (GC) used was unable to distinguish between H₂ and CH₄. The H₂-CH₄ compounds were detected during core sampling in the background gas ranging from 0.6 to 1.1 vol% (Text S8 in Supporting Information S1). The EDR makes no mention of hydrocarbons when crossing the Lusitanian formation.
 - The Hericy 1 well (drilled in 1986) detected H₂ and CH₄ in the gas background during mud gas logging. However, the GC could not distinguish between them, similar to Luteau 1. This potential H₂ detection occurred during intersection of the Lower Jurassic lithology composed of claystone. The gas background ranged from 1 to 3.5 vol% (Text S9 in Supporting Information S1).
 - The Saint Martin de Bossenay 17 well (drilled in 1976) detected H₂ traces at two different depths. (a) The Upper Triassic claystone interval was tested at a depth of 2,084 m. H₂ gas was detected, but its concentration was not reported. The well was open for 32 min, during which 20 L of gas and 0.5 L of mud were recovered. (b) The Upper Triassic lithology, composed of sandstone, claystone, and anhydrite rocks, was tested at a depth of 2,295 m. During the test, the well was open for 37 min and recovered 410 L of mud, a small quantity of gas and 120 L of gasified mud. It was also reported that the gas was detected in the same formation as the Saint Martin de Bossenay 201 well. The pressure of 196 kg/m² was unstabilized at 2,321 m (Text S10 in Supporting Information S1). The EDR mentions the presence of low-level fluorescence on excavated material.
 - The Jeumont 1 well (drilled in 1963) detected H₂ in mud gas at two different depths. (a) At a depth of 4,443 m in Upper Devonian, composed of quartzite and argillite. H₂ is present in background gas at a concentration of 0.5 vol%. (b) At a depth of 4,807 m, in Middle Devonian is mainly composed of quartzite and shale. H₂ is also detected in the background gas at a concentration of 1.8 vol%. The H₂ concentration for both depths was unexplained (Text S11 in Supporting Information S1).

All the data originate from wells drilled at various times during the twentieth century and these occurrences have not been previously correlated. To comprehend the natural hydrogen potential of this region, it is essential to integrate these data with the knowledge acquired in recent decades and with contemporary geological studies of the Paris Basin. Consequently, a comprehensive compilation of both geochemical and geological knowledge will also be carried out.

5. Discussion

Through the analysis of this extensive dataset, we have identified 26 H₂-bearing wells distributed across the French sedimentary basin (Figure 3, Table S2). These wells can be categorized into two distinct groups: those situated in the eastern region of Paris, and those in the southern part of the Aquitaine Basin.

In the Aquitaine Basin, the observed concentrations of H_2 can be linked to the geological context. Indeed, these wells are at the vicinity of a mantle body (peridotite) present at relatively shallow depths and are also located near major faults (e.g., major faults) facilitating fluid migration (Chevrot et al., 2018, 2022; Jammes et al., 2010; Lefeuvre et al., 2021, 2022; Lehujeur et al., 2021; Tugend et al., 2014). Thus, within the scope of this study, we will specifically focus on investigating the origin of the H_2 anomalies in the wells of the Paris Basin.

In the Paris Basin, the presence of H_2 -rich wells constitutes an enigma, as the occurrence of hydrogen cannot be justified by the local geology, which does not a priori present the conditions usually favorable for hydrogen production. Consequently, our study will focus on investigating the underlying geological factors influencing H_2 presence in the Paris Basin. Particular attention is paid to the artificial H_2 generation processes such as steel corrosion, drill bit metamorphism, or rock comminution (so called mechano-radicals) that may occur during drilling and bias the interpretations. Our findings reveal promising indications of H_2 potential in the Paris Basin. This intracratonic basin is characterized by a geologically diverse basement comprising peridotite rocks at relatively shallow depths (less than 4 km; Averbuch & Piromallo, 2012). Notably, the basin is intersected by major faults that extend through both the basement and the overlying sedimentary cover, which could be conduits for H_2 migration and accumulation.

5.1. Hydrogen Detection and Origin Hypotheses in Various Geological Formations of the Paris Basin

According to the OCR algorithm, the H_2 -bearing well locations are concentrated within an area of 8,600 km² in the middle east of the Paris Basin (Figure 1a). The H_2 concentration in the Paris Basin is not randomly distributed throughout all formations but is mainly present in three formations: Lusitanian, Dogger, and Upper Triassic.

5.1.1. The Lusitanian Aquifer

Three sites north of the Bray Fault—Longueil, Betz, and Cramaille—reported H_2 content while intersecting the Lusitanian aquifer. According to the initial interpretations, this was potentially due to tool degradation. However, H_2 was also detected in the same formation in the Longueil and Betz wells. The well geologist revealed in the Betz 101 EDR that detecting H_2 in the Lusitanian aquifer is common. Therefore, the hypothesis of H_2 production by steel corrosion in Cramaille can be discarded. An alternative hypothesis involves the artificial production of H_2 through mechano-radical reactions occurring during drilling. This suggests that the mechanical crushing of rocks results in the production of fresh silicate surfaces that are highly reactive with water and generate H_2 (Hirose et al., 2011; Kita et al., 1982). Nevertheless, the Lusitanian is mainly composed of limestone and contains little to no silica. Lefeuvre (2022) conducted a limestone grinding experiment in a confined atmosphere and demonstrated that this rock did not produce H_2 . This leads to questioning the origin of the H_2 . In the subsequent sections, we aim to address these questions comprehensively.

5.1.2. The Dogger Aquifer

Investigations into the Dogger aquifer, specifically at the Montreuil Aux Lions, Connantre, Longueil and Grandville wells north of the Bray Fault, have identified H_2 concentrations. At Montreuil Aux Lions, gas was directly sampled at depth within the Dogger aquifer using a pressure bottled introduced in the well. The analysis of this sample revealed 52 vol% H_2 concentration within a 1-L vessel under 25 bar pressure. Notably, the hypothesis of H_2 production by mechano-radical processes was discarded due to both the surprisingly high concentration of H_2 and the composition of the Dogger aquifer, which is an Oolitic limestone. Moreover, this conclusion is reinforced by the fact that the sampling procedure did not take place at the same time as the drilling activities.

Gases have been sampled in the Dogger aquifer from geothermal production wellheads during both artesian flow and production under pumping (Figure 2). These measurements at the wellhead exhibited pressure ranges from 4 to 12 bar, indicative of a monophasic state (e.g., dissolved gases) at depth for such fluids (Marty et al., 1988). Marty et al. (1988) conducted analyses on 34 dissolved gas samples, which were isolated using a vacuum flask vessel half-filled with the separated liquid phase containing the dissolved gases. They recorded H_2 concentrations reaching up to 12×10^{-5} mol/l or 24 ppm (at 71°C and 9.81 bar). The occurrence of steel corrosion as a possible origin for the elevated H_2 levels remains a matter of concern. However, only 9 out of the 34 wells displayed concentrations exceeding 1×10^{-5} mol/l, suggesting a different origin for the elevated H_2 concentrations. It's

important to highlight that in the samples collected, the concentration of methane is 10 times higher than that of hydrogen. These observations support the existence of an H_2 - CH_4 -rich aquifer in the Dogger aquifer.

The Dogger aquifer is also known for the presence of sulfate-reducing methane-producing bacteria, which are mainly thermophilic (Fouillac et al., 1990). Isolated methane-producing bacteria from Mellaray's well have shown the capability to thrive utilizing H_2 and CO_2 as their sole carbon and energy sources (Daumas et al., 1986; Marty et al., 1993). This observation suggests that the H_2 measured in the Dogger is a source of energy for bacterial communities and raises questions about the deeper origin of H_2 .

5.1.3. The Triassic Aquifer

In the Upper Triassic Formation situated south of the Bray Fault, The presence of H_2 was documented in the Le Luteau, Grandville, Longueil and Saint Martin de Bossenay wells. This formation is characterized by a composite of clay, anhydrite, and sandstones. At Le Luteau, the detected background H_2 concentrations during drilling (mud gas logging) might be associated with mechano-radical mechanisms, a hypothesis supported by the silica-rich composition of the rocks. At Saint Martin de Bossenay, H_2 detection occurred during post-drilling production testing, rendering the mechano-radical hypothesis less plausible for this site. It is important to note that no acidification of the well was performed prior to conducting this test. In this formation, the anhydrite may act as a promising sealing rock for trapping H_2 .

5.1.4. H_2 Detection Within the Basement

Lastly, the two wells at Coubert and Jeumont reached the basement, and detected H_2 at this depth. The Coubert well, situated to the south of the Bray Fault, crosses a geological formation composed predominantly of Gneiss and quartzite with galena inclusions. At this site, it was not possible to differentiate between CH_4 and H_2 gases due to instrumental reasons. However, a notable diminution in C4 to C2 hydrocarbon concentrations suggests possible H_2 degassing, potentially originating from drilling tools or the surrounding rock matrix as suggested in the EDR. The Jeumont well only revealed H_2 concentration during core sampling of the quartzite. This observation supports the hypothesis of H_2 generation through mechano-radical processes.

Due to limited EDR data, deciphering the source of H_2 remains challenging. Geological and geochemical contextualization may offer a starting point for further analysis.

5.2. Geological Trends and Structural Analysis of Drilling Wells in the Paris Basin

Initial observations indicate that the wells Longueil 1, Betz 101, and Montreuil Aux Lions 1 located to the north of the Bray Fault are aligned to a trend of $\text{N}130^\circ$ orientation. This aligns with the orientation of the Bray Fault, albeit situated approximately 20 km northward. This trend is highlighted and well correlated with the map of the vertical gradient of the pole-reduced magnetic anomaly extended to 600 m (Figure 4; Baptiste et al., 2016). The vertical gradient technique can be used to identify lithological features, delineate contacts and discern structural variations between distinct geological formations (Baranov, 1953). This pattern could correspond to the Bray Thrust located by Averbuch and Piromallo (2012), but regarding the seismic cross-section, this discontinuity does not affect the sedimentary cover (Figure 1b).

In contrast, the Cramaille 101, Connantre 1, 2, and Grandville 109 wells do not align with this specific trend but are situated within a large regional anomaly. This anomaly has been linked to magmatic rocks, with the axis of this anomaly suggesting a correlation with the closure of the Rhenohercynian ocean during the Carboniferous period (Thébault et al., 2006).

Regarding the wells located south of the Bray Fault (Figure 4), their distribution is more random and their association with structural features is less obvious. These wells are part of the Bloc Every-Tonnerre, an area where the basement has been mapped through magnetic and gravimetric analysis (Baptiste, 2016).

The Hericy 1 and Saint Martin de Bossenay 17 wells are located on the Evry Fault (Baptiste, 2016). The mapped anomalies are due to the presence of basic rocks along the Evry fault, and a combination of gneissic or granitic rocks interspersed with Néoproterozoic/Paleozoic. The two other wells, Le Luteau 1 and Coubert 1, while not associated with any specific faults, are located above the granodiorite basement.

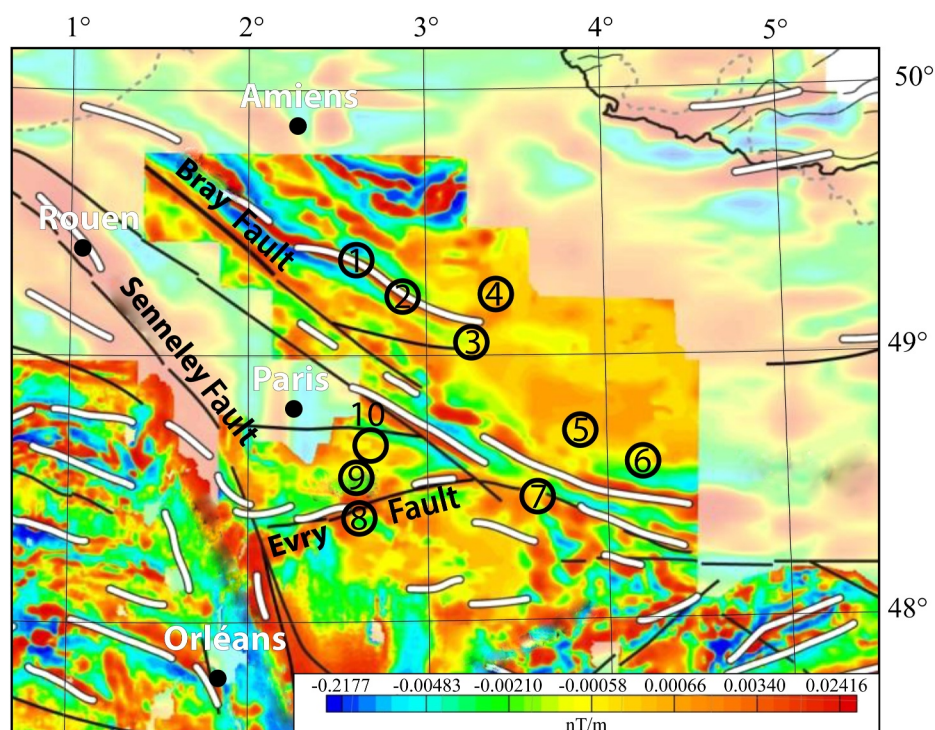


Figure 4. Map of the vertical gradient of the magnetic anomaly reduced to the pole and extended to 600 m up. Black lines correspond to major discontinuities, white lines to magnetic axes and the black circle corresponds to the H₂-bearing well 1—Longueil; 2—Betz; 3—Montreuil Aux Lions; 4—Cramaille; 5—Connantre 1 and 2; 6—Grandville; 7—Saint Martin de Bossenay; 8—Hericy; 9—Le Luteau; 10—Coubert (Modified from Baptiste (2016)).

While the majority of the wells' locations appear to be correlated with fault lines, further investigation is needed to determine if these faults could act as conduits for fluid migration.

5.3. Characteristics and Deep Fluid Helium Studies of the Dogger Aquifer in the Paris Basin

The Paris Basin is characterized by several key aquifers, including Triassic sandstones, Dogger limestones, Lusitanian limestones, and Lower Cretaceous sandstones (Figure 2). Among these, the Dogger aquifer has been the primary focus of numerous studies, primarily due to its notable helium concentration (Bril et al., 1994; Castro, Goblet, et al., 1998; Castro, Jambon, et al., 1998; Marty et al., 1988, 1993; Pinti & Marty, 1998).

The Dogger aquifer exhibits a relatively consistent helium concentration and isotope ratio though it presents minor variations in water chemistry (Marty et al., 1993). According to Marty et al. (1988), the reported dissolved gas concentration of helium in the aquifer ranges from 1.02 to 4.65×10^{-5} mol/l. Additionally, isotopic analyses have revealed a distinct excess of ³He, indicative of a mantle-derived origin. Pinti and Marty (1998) suggest that the enrichment in mantle-derived ³He, along with high ⁴He concentrations, may be attributed to the influx of basement fluids into the Dogger and potentially combined with in situ production within the Dogger.

To explain the high helium concentration in the aquifer, one plausible explanation is the presence of a preferential migration pathway, notably the Bray-Vittel Fault (Figure 2). This fault affects both the sedimentary cover and the basement, and can serve as intermittent vertical conduits for helium-rich fluids through the 700 m of low-permeability rocks that separate these aquifers (Bril et al., 1994; Pinti & Marty, 1998; Worden & Matray, 1995).

The analysis of the Dogger aquifer has revealed fluid migration along the major faults, while our OCR analysis has pinpointed wells exhibiting H₂ anomalies correlated with these faults. These findings suggest the existence of a potentially fertile H₂ system.

5.4. H₂ Formation: Organic Matter and Coal in the Paris Basin

Organic matter diagenesis may generate small amounts of H₂, but this gas has never been documented in significant concentrations in natural gas reservoirs (Milkov, 2022). However, H₂ is frequently reported in coal bed methane (Boreham et al., 2023; Levshounova, 1991; Mahlstedt et al., 2022). Recent investigations in Lorraine (France) have reported elevated dissolved H₂ and CH₄ concentrations in a flooded well formerly drilled within the carboniferous formation to produce coal bed methane (Allouti et al., 2023). These observations reinforce the rationale to seriously consider coal formations as potential sources for H₂ in the Paris Basin. Unfortunately, the EDR reports investigated in this study do not mention the presence of coal in the wells where H₂ was reported. This precludes a more detailed analysis of the connection between H₂ concentration anomalies and coal formation, even if this point deserves a more specific attention.

5.5. A Putative H₂ System?

The potential for H₂ exploration in the Paris Basin is closely linked to its geodynamic characteristics and structural features, particularly those associated with the closure of the Rhenohercynian ocean. This suture zone is composed of ultramafic rocks, including peridotites and amphibolites, which are able to produce H₂ through hydrothermal reactions such as serpentinization (Figure 2). The current temperature near this discontinuity is approximately 120°C (Pinti & Marty, 1998), significantly lower than the 250–300°C typically required for optimal H₂ production kinetics (Malvoisin et al., 2012; McCollom, 2016; McCollom & Donaldson, 2016). However, a significant aspect of the Lizard complex composition is that the dunite is largely serpentinized and contains magnetite (Leake & Styles, 1984). Recent findings by Geymond et al. (2023) have revealed the potential of magnetite to facilitate H₂ production at relatively lower temperatures through hydrothermal alteration processes. Therefore, in this geological context, serpentinized dunite, particularly rich in magnetite, may represent a viable source rock for H₂ generation. This raises the question of the actual production mechanism of H₂ in this area.

To the north of this closure zone, magnetic and gravimetric anomalies suggest the presence of granitic rocks, which can facilitate H₂ generation via radiolysis of water, a process independent of rock temperature (Lin, Hall, et al., 2005; Sherwood Lollar et al., 2006).

A key factor in these H₂-producing processes is the presence of a water source, originating from a recharge zone. The basin's sedimentary cover and basement are influenced by regional tectonic lineaments, notably the Bray-Vittel and Rouen-Couy faults, which may act as conduits for vertical fluid migration. Studies on noble gases suggest a vertical flow of fluids through the Rouen-Couy fault, allowing water infiltration into the sedimentary layer through this discontinuity, which eventually reaches the basement (Pinti & Marty, 1998; Rouchet, 1981). Additionally, the Paris Basin hosts multiple aquifers, potentially serving as sources for water essential to these reactions.

Helium analysis within the basin has provided insights into fluid migration patterns, with mantle-derived helium detected in the Dogger aquifer, as evidenced by the excess of ³He. H₂ produced in the basement may migrate through the Bray-Vittel fault and major detachments to reach the aquifer. The Triassic and Jurassic Formations contain evaporitic and clayey rocks, offering effective sealing properties for H₂ entrapment. Furthermore, the temperature in the Triassic Formation seems suitable for H₂ trapping over geological time, being too hot for microbial activity and too cold for efficient abiotic reactions involving H₂ (Lefevre et al., 2022).

The Paris Basin stands as a good case study for natural H₂ exploration. All the essential components necessary for the establishment of a H₂ system are gathered: (a) a water source facilitating H₂ production, (b) iron-rich and granitic rocks, (c) preferential migration pathways along faults connecting both the basement and sedimentary layers, and (d) efficient reservoirs characterized by the presence of claystone, evaporites, and aquifers, which may act as effective seals.

6. Conclusions

After processing well reports in the CVAGeoDB in-house database with an OCR algorithm, we discovered multiple wells where H₂ was detected. This constitutes the first use of such a methodology on a drilling database in France, resulting in the identification of wells with reported H₂ concentrations in their End-of-Drilling Reports

(EDRs). The distribution of these wells is not random in terms of geological context; they are frequently situated near mantle bodies or ophiolitic rocks at shallow depths. This discovery should lead to further research.

In the Paris Basin case, the OCR algorithm revealed four main formations in which H₂ has been detected: Lustanien, Dogger, Triassic aquifers, and basements. A maximum of 52 vol% of H₂ was measured at depth (in-situ sampling) in the Dogger, whereas up to 6 vol% was reported in a well crosscutting the Lusitanian limestone. In the other formations, the concentration of H₂ was not measured, but its presence was still reported. These wells are situated in a small area in the central-east of the Paris Basin, and their distribution does not seem random. All of them are situated along the Bray Fault, which is a N°130 fault that affects both the basement and sedimentary cover. The basement comprises rocks that have the potential to be sources of H₂, such as peridotite or granitic rocks. An excess of ³He is reported in the Dogger aquifer, suggesting a contribution from the mantle and deep fluid migration. Finally, evaporite and clay-bearing formations reported in the basin may provide a promising seal over traps for H₂.

Data Availability Statement

Our analysis is based on public data extracted from an internal database archived by the CVA group and can be found in Supporting Information S1. Data can be accessed via <https://www.minergies.fr/fr/page/acces-aux-donnees>. Simply enter the associated borehole number (Table S1) and you will have access to the borehole geological log and the EDR. The OCR python library can be accessed via this link: <https://github.com/madmaze/pytesseract>.

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